# Seong Hyeon Kim

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## Education

**Pohang University of Science and Technology (POSTECH)**, Ph.D. in Artificial Intelligence — Feb 2026 – Present
- Advisor: Prof. Hyemin Ahn | Lab: Robot Learning Lab (RLLab)

**Ulsan National Institute of Science and Technology (UNIST)**, M.S. in Artificial Intelligence — Mar 2024 – Feb 2026
- Advisor: Prof. Hyemin Ahn | Lab: AI-HRI Lab

**Myongji University**, B.S. in Mechanical Engineering — Mar 2018 – Feb 2024

## Experience

**SuperbAI**, Research Intern — Dec 2025 – Feb 2026
- Project: **Text-to-Humanoid Control**
- Developed motion synthesis algorithms for humanoid robots conditioned on natural language.

## Research Experience

**Graduate Student Researcher**, Robot Learning Lab (RLLab) – POSTECH — Feb 2026 – Present

**Graduate Student Researcher**, AI-HRI Lab – UNIST — Mar 2024 – Feb 2026
- Focusing on **Robotic Manipulation** and **World Models** for intelligent robot control.

**Undergraduate Intern**, Dynamic Robot Control Lab (DRCL) – Myongji University — Mar 2019 – Feb 2024
- Conducted research on mechanical design and parallel mechanisms for wheel-legged robots.

## Publications

**Proprioception-conditioned Visual Scene Generation for Robot World Modeling via Contrastive Learning and Diffusion** — 2025
**S. H. Kim**, H. Ahn | *Journal of ICROS*

**Design of a Wheel-Legged Robot using Linear 2-DOF Parallel Mechanism** — 2023
**S. H. Kim**, D. Choi | *KSME Conference*

**Application of TDOA Algorithm for Sound Source Direction Tracking** — 2023
T. Kim, **S. H. Kim**, D. Choi | *KSME Conference*

**2-DOF Parallel Mechanism for Linear Wheel-Legged Robot System** — 2023
**S. H. Kim**, T. Kim, S. Ryu, J. Han, D. Choi | *KSME Conference*

**Ball Tracking and Trajectory Measurement System for Tennis Robot** — 2022
Y. Yang, S. Ahn, **S. H. Kim**, D. Choi | *KSME Conference*

**Real-time Object Trajectory Prediction using AI and Stereo Vision** — 2022
Y. Yang, S. Ahn, **S. H. Kim**, D. Choi | *KSME Conference*

## Honors & Awards

**Bronze Award**, School Capstone Design Project – Myongji Univ. (Elderly Companion Robot) — Dec 2023
**Bronze Award**, Hanium ICT Mentoring – KFII (Elderly Companion Robot) — Dec 2023
**Silver Award**, SEP Convergence Festival – Myongji Univ. (Elderly Companion Robot) — Dec 2023
**Encouragement Award**, Capstone Design Competition – KSME (Sound Source Tracking) — Apr 2023

**Best Paper Award**, Spring Conference – KSME (Tennis Ball Tracking) Apr 2023
**Bronze Award**, Capstone Design Competition – Myongji Univ. (Desert Seeding Robot) Dec 2019

## Professional Service

**Reviewer**, IEEE International Conference on Robotics and Automation (ICRA) 2025
**Volunteer**, LG Project-based Learning Program Summer 2025

## Technical Skills

**Programming:** Python, C, MATLAB

**Tools:** ROS/ROS2, MuJoCo, Gazebo, PyTorch, Git, SolidWorks, KiCad, LaTeX

**Robots:** Unitree G1, Unitree Go2, Universal Robots (UR3, UR5), Franka Emika Panda